Experiment #0: Computational Chemistry

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Section: 001

Date Performed: 2026-02-03
Date Submitted: 2026-02-17

Introduction

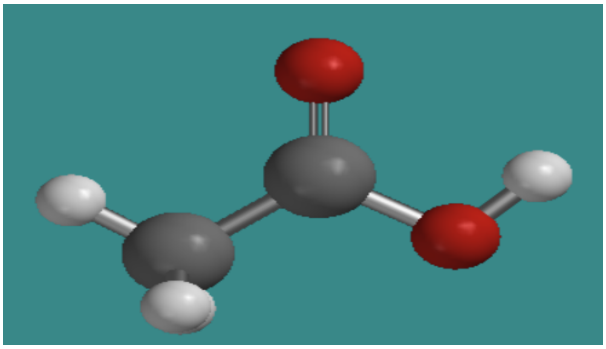
The chemical and physical properties of molecules can be explained and predicted by the dynamical states of quantum systems determined by solutions to the Schrödinger equation.[1] However, due to the high degrees of freedom in many-body systems of interacting molecular nuclei and electrons, exact analytical solutions to the Schrödinger equation are unavailable, often restricted to small systems with single electron. Consequently, specialized computational algorithms must rely on systematic approximations to yield tractable solutions for complex molecules. The purpose of this experiment is to utilize the *Spartan*[2] software suite to investigate molecular properties, optimized geometries, dipole moments, and vibrational normal modes for acetic acid and formaldehyde.

Within the framework of quantum mechanics, the foundational simplification in molecular quantum mechanics is the Born-Oppenheimer approximation.[3] Because electrons have a significantly smaller mass than nuclei while experiencing forces of comparable magnitude, their velocities are vastly greater. This allows electronic and nuclear motions to be effectively decoupled adiabatically. It can be reasonably assumed that the electrons instantaneously adapt to a given nuclear configuration, allowing the electronic Schrödinger equation to be solved for fixed nuclear coordinates to map out a potential energy surface and tremendously simplify the computations.

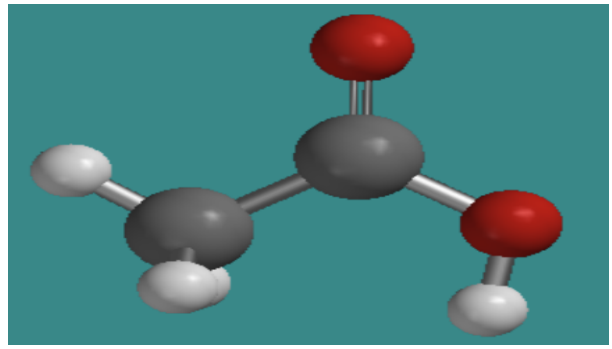
To compute the electronic structure of target molecules, this experiment utilizes the Hartree-Fock[4,5,6] (HF) method with 3-21G[7] molecular orbital basis set. HF theory is a variational approach that simplifies the many-electron problem by invoking the independent particle approximation, wherein electron-electron repulsion is treated only in an average way. And the 3-21G basis set uses a split-valence representation to model the molecular orbitals. Because the operator in the Hartree-Fock eigenvalue equation depends on its own eigenfunctions, the equations are inherently non-linear and must be solved iteratively using the self-consistent-field (SCF) method. In this experiment, these computational methods are applied to determine the optimized equilibrium geometries of the target molecules, allowing for the subsequent calculation and analysis of their electric dipole moments and infrared vibrational spectra.

Results

The optimized geometries of the two acetic acid conformers are presented in Figure 1. Table 1 summarizes their calculated parameters, including dihedral angles, strain energies, point groups, and dipole moments obtained by molecular mechanics (MMFF) and HF/3-21G optimizations. For the more stable conformer (Conformer 1), the detailed bond lengths and bond angles are recorded in Table 2 and Table 3 respectively.



(a) Conformer-1



(b) Conformer-2

Figure 1: Optimized Conformers of Acetic Acid

Table 1: Calculated Parameters of Acetic Acid Conformers

Parameter	**Conformer-1**	**Conformer-2**
Dihedral angle (H-O-C=O)	0.00°	0.00°
Strain energy (kcal/mol)	-110.4824	-85.9140
Point group	C_s	C_s
Symmetry elements present	E, σ	E, σ
Conformation Eclipsed/staggered	Eclipsed	Eclipsed
Dipole moment (D)	1.65	4.97
Methyl group H-C-H angles (°)	108.63, 108.63, 110.61	108.55, 108.55, 110.70

For formaldehyde, the geometry optimization and IR spectrum calculations were performed at the HF/3-21G level. The optimized structure is shown in Figure 2, with a C_{2v} symmetry and the optimized energy and molecular orbital energies are in Table 4. The visualized molecular orbital surfaces (LUMO+1, LUMO, HOMO, HOMO-1) are presented in Figure 3. The calculated infrared spectrum is displayed in Figure 4. Table 5 compares the calculated vibrational frequencies against experimentally observed values, and the atomic motions for these six normal modes are illustrated in Figure 5.

Table 2: Bond Lengths for Acetic Acid (Conformer 1)

Bonds	**Bond length (Å)**
C=O	1.219
C–C	1.493
C–H	1.094
C–O	1.346
O–H	0.980

Table 3: Bond Angles for Acetic Acid (Conformer 1)

Bonds	**Bond angle (°)**
H–C–H	108.63 / 110.61
H–C–C	109.82
C–C=O	126.56
O–C=O	121.02
C–O–H	104.05

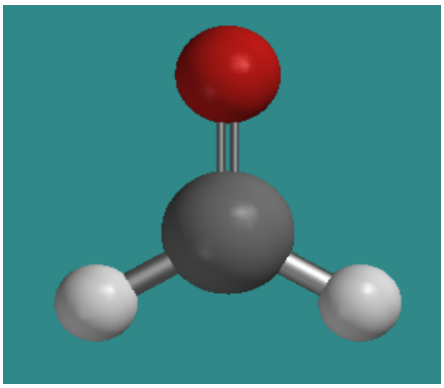


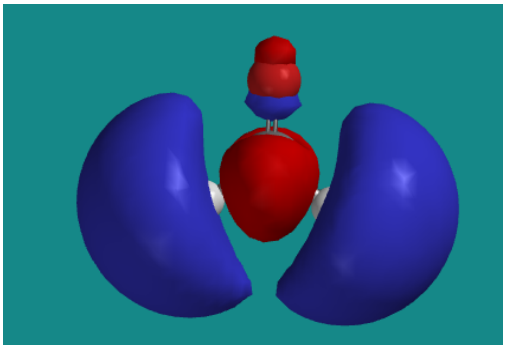
Figure 2: Optimized Structure of Formaldehyde

Table 4: Calculated Parameters of Formaldehyde (HF/3-21G)

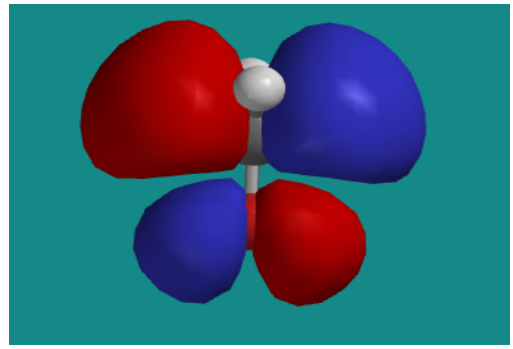
Parameter	**Value**
Optimized energy (kcal/mol)	0.2266
Calculated vibrational modes	6
LUMO energy (eV)	4.01
HOMO energy (eV)	-11.78
HOMO-LUMO gap (eV)	15.79

Table 5: Experimental and Calculated Infrared Frequencies (cm^{-1}) of Formaldehyde

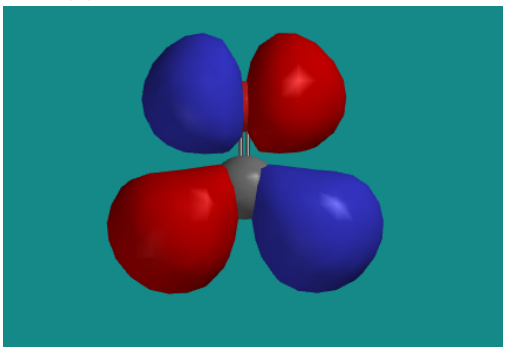
Symmetry	**Description of the mode**	**Observed**	**Calculated**	**Calculated (scaled 0.9)**
A_1	Symmetric CH_2 stretching	2783	3162	2845.8
A_1	CO stretching	1746	1915	1723.5
A_1	CH_2 scissoring	1500	1693	1523.7
B_1	CH_2 asymmetric stretch	2843	3233	2900.7
B_1	CH_2 rocking	1249	1379	1241.1
B_2	Out of plane bend	1167	1337	1203.3



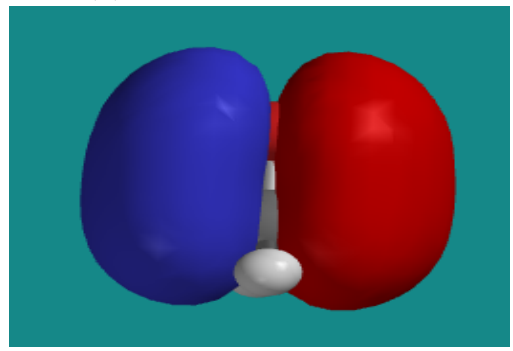
(a) Visualization of LUMO+1



(b) Visualization of LUMO



(c) Visualization of HOMO



(d) Visualization of HOMO-1

Figure 3: Molecular Orbitals of LUMO+1, LUMO, HOMO, and HOMO-1

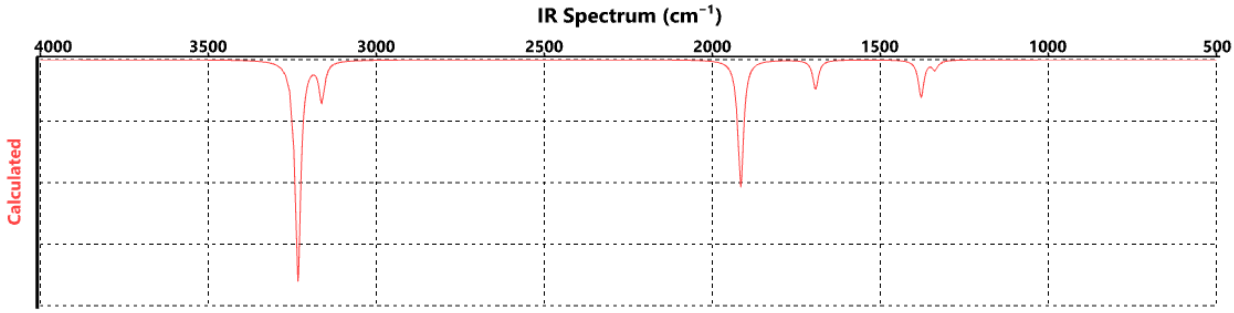
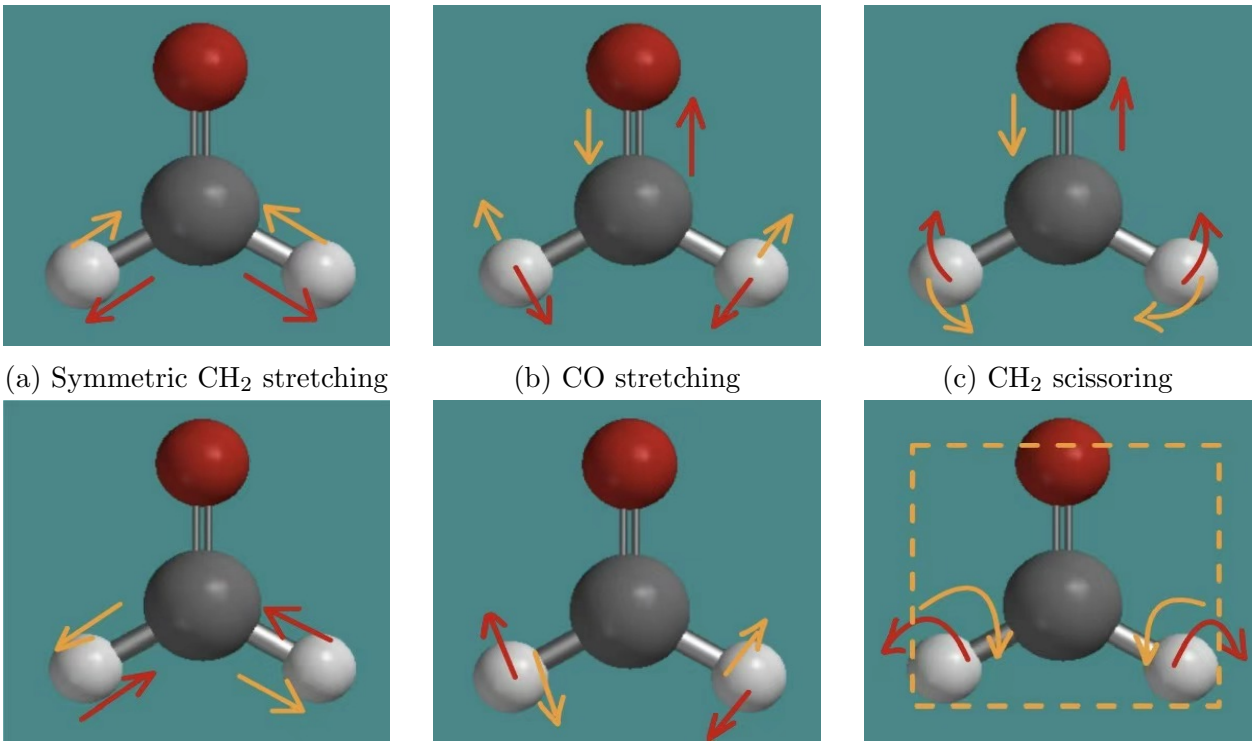


Figure 4: Calculated Infrared Spectrum of Formaldehyde



(a) Symmetric CH_2 stretching (b) CO stretching (c) CH_2 scissoring

(d) CH_2 asymmetric stretch (e) CH_2 rocking (f) Out of plane bend

Figure 5: Six Normal Vibrational Modes of Formaldehyde

Discussion

As presented in the results for acetic acid, both conformers have C_s symmetry, with the carbonyl carbon, carbonyl oxygen, hydroxyl carbon, hydroxyl oxygen, methyl carbon, and one methyl hydrogen on the symmetric plane. When evaluating their relative stability, conformer 1 exhibits a lower strain energy (-110.4824 kcal/mol) than conformer 2 (-85.9140 kcal/mol), indicating conformer 1 as the more stable configuration. The reason for the stability is that conformer 1 minimizes electrostatic repulsion by opposing the bond dipoles of the C=O and O-H groups, whereas conformer 2 aligns them. Moreover, the orientation of the C=O bond and the in-plane methyl hydrogen is eclipsed in both conformers. The eclipsed conformation is favored because it can stabilize the hyperconjugation effects between the filled σ_{C-H} orbital and the empty $\pi^*_{C=O}$ orbital. Furthermore, while the three H-C-H angles in the methyl group are not strictly equal due to conformational influences, the exact equivalence of two of these angles (108.63°) is consistent with the existence of the single plane of symmetry within the molecule.

Comparing the HF/3-21G calculated dipole moment of conformer 1 (μ = 1.65 D) to the experimental literature value ($\mu = 1.70 \pm 0.03$ D[8]), the calculation proves to be highly accurate with an error rate of merely 2.94%. The difference in dipole moments between the two conformers (1.65 D and 4.97 D) originates directly from the relative spatial orientation of the polar C=O and O-H bonds. In conformer 1, the O-H bond is oriented away from the carbonyl oxygen, causing their respective bond dipole vectors to point in opposing directions and partially cancel each other out. Conversely, in conformer 2, the O-H bond points toward the carbonyl oxygen, aligning the bond dipoles such that they reinforce one another. The alignment generates a significantly larger net molecular dipole moment and increases the intramolecular electronic repulsion between the partially negative oxygen atoms, making the conformer 2 unstable.

As for the formaldehyde molecule, a non-linear system containing $N = 4$ atoms is expected to have $3N - 6 = 6$ vibrational normal modes, which perfectly matches the six modes generated by the Spartan calculation. As summarized in Table 5, the raw calculated frequencies systematically overestimate the true harmonic frequencies and applying the standard 0.9 empirical correction factor brings the HF theory calculations into precise agreement with the experimentally observed values. Furthermore, the symmetry labels assigned to these modes perfectly characterize the visual animations in Figure 5. The A_1 modes are totally symmetric and remain invariant to operations, while the B_1 and B_2 modes behave anti-symmetrically.

Finally, evaluating the molecular orbital surfaces (Figure 3) reveals different localizations of electron density. The HOMO-1 has a single nodal plane running through the molecule. The orbital has no additional node between the C and O nuclei, consistent with a π bonding combination. The HOMO surface exhibits positive and negative lobes primarily localized on the non-bonding p-orbitals of the oxygen atom, with a nodal plane mapping through the nuclei. The LUMO exhibits π^* anti-bonding character with a clear node situated explicitly between the carbon and oxygen atoms. The LUMO+1 has a nodal plane passing through the carbon–oxygen axis causes a phase change from the large blue lobes on the sides to the red lobe at carbon. Meanwhile the plane containing C, O, and H also serves as a nodal plane, creating separation at oxygen where the wavefunction exhibits opposite phases. The difference among HOMO-1, HOMO, LUMO, and LUMO+1 surfaces shows the fundamental

quantum mechanical trend that as the energy of the molecular orbital increases, the number of spatial nodes correspondingly increases, which leads to a decrease in the degree of bonding overlap of the electron wavefunction between atoms.

References

[1] Henry C. Brenner, D. S. J. W. A. H. N. K. N. E. N. M. K. P. S. C. S., Pance Naumov Laboratory Manual of Physical Chemistry. 2015; Experiment 10: Computational Chemistry.

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[3] Born, M.; Oppenheimer, R. Zur quantentheorie der molekeln. *Annalen der physik* **1927**, *389*, 457–484.

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[5] Fock, V. Näherungsmethode zur Lösung des quantenmechanischen Mehrkörperproblems. *Zeitschrift für Physik* **1930**, *61*, 126–148.

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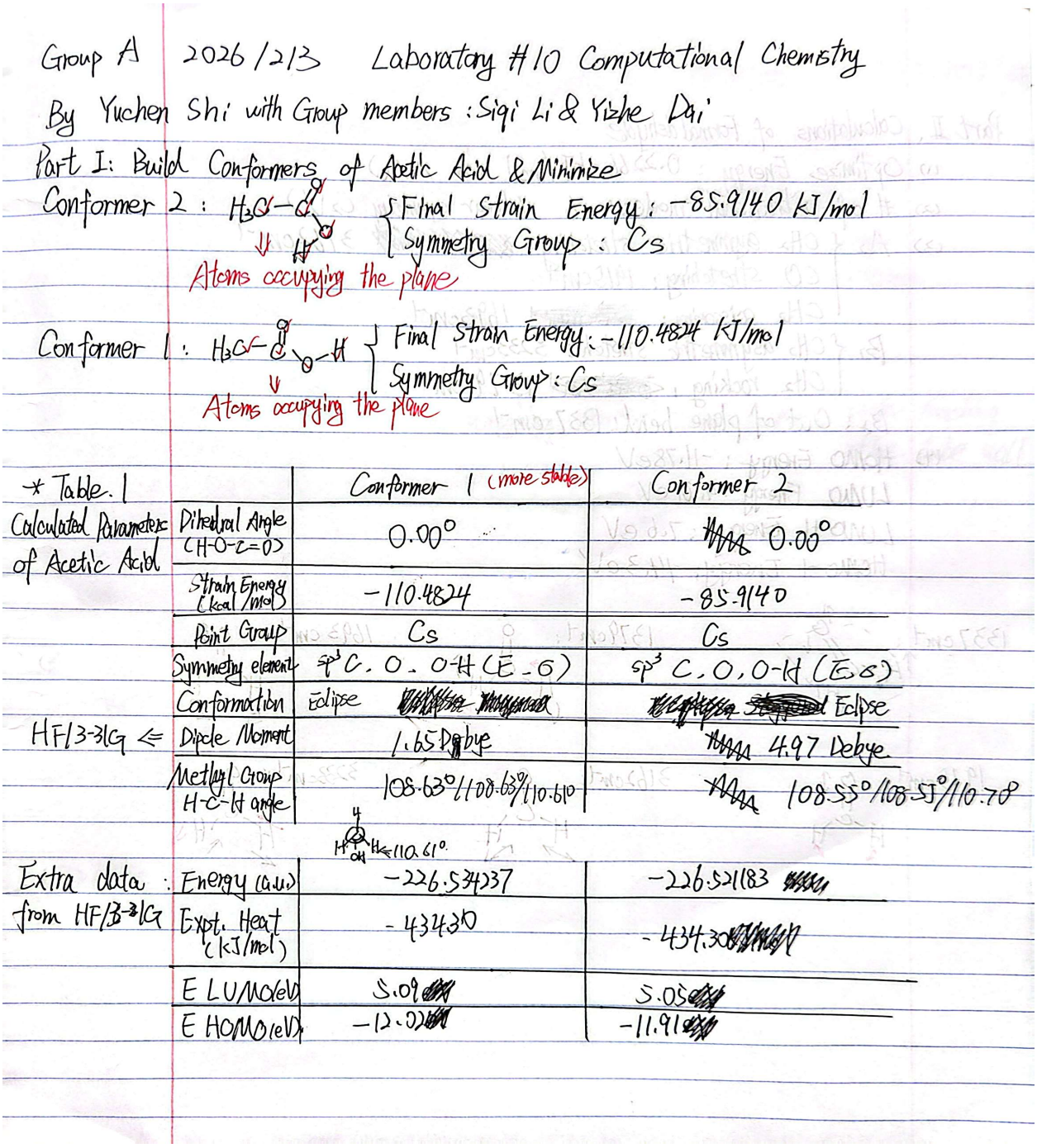
[8] Lide, D. *Handbook of Chemistry and Physics; CRC Press: Boca Raton, FL*; 2002.

Appendices



Group A 2026/2/3 Laboratory #10 Computational Chemistry

By Yuchen Shi with Group members: Siqi Li & Yizhe Dai

Part I: Build Conformers of Acetic Acid & Minimize

Conformer 2: Final Strain Energy: −85.9140 kJ/mol; Symmetry Group: Cs

Atoms occupying the plane

Conformer 1: Final Strain Energy: −110.4824 kJ/mol; Symmetry Group: Cs

Atoms occupying the plane

* Table. 1 Calculated Parameters of Acetic Acid		Conformer 1 (more stable)	Conformer 2
	Dihedral Angle (H-O-C=O)	0.00°	0.00°
	Strain Energy (kcal/mol)	−110.4824	−85.9140
	Point Group	Cs	Cs
	Symmetry element	sp^3 C, O, O-H (E, σ)	sp^3 C, O, O-H (E, σ)
	Conformation	Eclipse	Eclipse
HF/3-31G ⇐	Dipole Moment	1.65 Debye	4.97 Debye
	Methyl Group H-C-H angle	108.63°/108.63°/110.61°	108.55°/108.55°/110.70°
Extra data from HF/3-31G	Energy (a.u.)	−226.534237	−226.521183
	Expt. Heat (kJ/mol)	−434.30	−434.30
	E LUMO (eV)	5.09	5.05
	E HOMO (eV)	−12.02	−11.91

Figure 6: Lab Notebook

Lab 0 Computational Chemistry

1. Error rate of Dipole Moment in Acetic Acid

$$\text{error} = \frac{|1.65-1.70|}{1.70} \times 100\% = \frac{0.05}{1.70} \times 100\% \approx 2.94\%$$

2. HOMO-LUMO Energy Gap in Formaldehyde

$$4.01\,eV - (-11.78\,eV) = 15.79\,eV$$

3. Calculation of Vib-Freq

$3162 \times 0.9 = 2845.8\ cm^{-1}$

$1915 \times 0.9 = 1723.5\ cm^{-1}$

$1693 \times 0.9 = 1523.7\ cm^{-1}$

$3233 \times 0.9 = 2900.7\ cm^{-1}$

$1379 \times 0.9 = 1241.1\ cm^{-1}$

$1337 \times 0.9 = 1203.3\ cm^{-1}$

Figure 7: Post-Lab Computation Notes

```
SPARTAN STUDENT MECHANICS PROGRAM:   (Win/64b)                  Release  9.0.2


 Frequency Calculation

  Adjusted 1 (out of 24) low frequency modes

  Reason for exit: Successful completion
  Mechanics CPU Time :        0.313
  Mechanics Wall Time:       0.0367

SPARTAN STUDENT Quantum Mechanics Driver:  (Win/64b)     Release  9.0.2

Job type: Geometry optimization.
Method: RHF
Basis set: 3-21G(*)
Number of basis functions: 44
Number of electrons: 32
Parallel Job: 4 threads

SCF model:
 A restricted Hartree-Fock SCF calculation will be
 performed using Pulay DIIS + Geometric Direct Minimization


Optimization:
         Step      Energy          Max Grad.      Max Dist.
            1     -226.530998       0.023925       0.090000
            2     -226.534087       0.006110       0.019039
            3     -226.534222       0.001602       0.003780
            4     -226.534235       0.000711       0.001863
            5     -226.534237       0.000098       0.000511

  Reason for exit: Successful completion
  Quantum Calculation CPU Time :         8.84
  Quantum Calculation Wall Time:         2.64


SPARTAN STUDENT Properties Program:  (Win/64b)                 Release  9.0.2
  Use of molecular symmetry enabled

                     Cartesian Coordinates (Angstroms)
       Atom            X             Y             Z
    ---------    ------------- ------------- -------------

  1 H  H1          -1.1507274     0.0000000     1.3712829
  2 C  C1          -0.1566600     0.0000000     0.9546726
  3 H  H2           0.3898921     0.8742836     1.2848353
  4 H  H4           0.3898921    -0.8742836     1.2848353
  5 C  C2          -0.2364501     0.0000000    -0.5407814
  6 O  O1          -1.2278098     0.0000000    -1.2197714
  7 O  O2           1.0107492     0.0000000    -1.0834003
  8 H  H3           0.9811137     0.0000000    -2.0516730

  Point Group = CS  Order =  1 Nsymop =  2

  Closed-Shell Molecular Orbital Coefficients
  MO:                    1          2          3          4          5
  Eigenvalues:     -20.49810  -20.45371  -11.33929  -11.18467   -1.46805
          (ev)    -557.78185 -556.57409 -308.55803 -304.35037  -39.94774

                       A'         A'         A'         A'         A'
   1 H1    S        -0.00091    0.00086   -0.00121    0.00019    0.00549
   2 H1    S'       -0.00007    0.00266    0.00214    0.01472   -0.00746
   3 C1    S1       -0.00004   -0.00003   -0.00162    0.98751   -0.01780
   4 C1    S        -0.00058   -0.00023   -0.00290    0.10235    0.01900
   5 C1    PX       -0.00013   -0.00037    0.00010    0.00015   -0.00021
   6 C1    PY        0.00000    0.00000    0.00000    0.00000    0.00000
   7 C1    PZ        0.00047   -0.00015    0.00066   -0.00030   -0.02210
```

Figure 8: HF/3-21G Calculation of Acetic Acid (Conformer 1)

```
SPARTAN STUDENT Quantum Mechanics Driver:  (Win/64b)     Release  9.0.2

Job type: Geometry optimization.
Method: RHF
Basis set: 3-21G(*)
Number of basis functions: 22
Number of electrons: 16
Parallel Job: 4 threads

SCF model:
 A restricted Hartree-Fock SCF calculation will be
 performed using Pulay DIIS + Geometric Direct Minimization


Optimization:
         Step      Energy          Max Grad.      Max Dist.
            1     -113.221083       0.013759       0.048675
            2     -113.221818       0.001135       0.002722
            3     -113.221820       0.000132       0.000157

<step 2>
Job type: Frequency calculation.
Method: RHF
Basis set: 3-21G(*)

  Reason for exit: Successful completion
  Quantum Calculation CPU Time :         7.25
  Quantum Calculation Wall Time:         2.31


SPARTAN STUDENT Properties Program:  (Win/64b)                 Release  9.0.2
  Use of molecular symmetry enabled

                     Cartesian Coordinates (Angstroms)
       Atom            X             Y             Z
    ---------    ------------- ------------- -------------

  1 C  C1           0.0000002     0.0000001     0.0105546
  2 H  H2          -0.0000001     0.9133781     0.5929316
  3 H  H3          -0.0000001    -0.9133780     0.5929313
  4 O  O1          -0.0000001    -0.0000002    -1.1964175

  Point Group = CNV Order =  2 Nsymop =  4

  Closed-Shell Molecular Orbital Coefficients
  MO:                     1          2          3          4          5
  Eigenvalues:     -20.48363  -11.28089   -1.41771   -0.86980   -0.69930
          (ev)    -557.38809 -306.96876  -38.57795  -23.66852  -19.02900

                        A1         A1         A1         A1         B1
   1 C1    S1        0.00019    0.98639   -0.11970   -0.17197   -0.00000
   2 C1    S         0.00059    0.09356    0.13295    0.20095    0.00000
   3 C1    PX       -0.00000   -0.00000   -0.00000   -0.00000    0.00000
   4 C1    PY        0.00000   -0.00000   -0.00000   -0.00000    0.35655
   5 C1    PZ       -0.00249   -0.00136   -0.15494    0.16236    0.00000
   6 C1    S'        0.01784   -0.04555    0.13091    0.49521    0.00000
   7 C1    PX'      -0.00000    0.00000   -0.00000   -0.00000    0.00000
   8 C1    PY'      -0.00000    0.00000    0.00000   -0.00000    0.28183
   9 C1    PZ'      -0.01391   -0.00804    0.04315    0.13220    0.00000
  10 H2    S        -0.00077   -0.00197    0.02671    0.17336    0.16865
  11 H2    S'        0.00057    0.01360   -0.01765    0.05663    0.09963
  12 H3    S        -0.00077   -0.00197    0.02671    0.17336   -0.16865
  13 H3    S'        0.00057    0.01360   -0.01765    0.05663   -0.09963
  14 O1    S1        0.98353    0.00004   -0.21428    0.09500   -0.00000
  15 O1    S         0.09799    0.00066    0.20641   -0.09260   -0.00000
  16 O1    PX        0.00000    0.00000    0.00000   -0.00000    0.00000
  17 O1    PY        0.00000   -0.00000    0.00000   -0.00000    0.27920
  18 O1    PZ        0.00450   -0.00061    0.12955    0.10004   -0.00000
  19 O1    S'       -0.04768   -0.00056    0.66078   -0.36135    0.00000
```

Figure 9: HF/3-21G Calculation of Formaldehyde